

arbitrageur would go long the replicating portfolio and short the index, closing out once profits have been realized. A preponderant form of index arbitrage is basis trading where mispricing dynamics (difference between the spot-price and related futures contract) are exploited by buying the index and selling related futures contracts. Intuitively, the gain from one of these positions will exceed the loss on the other, resulting in a net gain.

Realizing the mispricing correction is in effect mean-reversion. Alexander (1999) details a cointegration based index arbitrage strategy to deliver stable returns. Following a cointegration statistical portfolio trading process:

1. We choose to go long-short in eight different countries with the European, Asian and Far East (EAFE) Morgan Stanley index as our benchmark. We aim to beat EAFE and are tasked with finding the basket of eight countries which are currently most highly correlated with the EAFE index.
2. Perform cointegration regression of log-EAFE price index y on log price indices of the local currencies $x_1, \dots, x_n$:

$$y_t = a + \beta_1 x_{1,t} + \dots + \beta_n x_{n,t} + \varepsilon_t \quad (13.27)$$

to find optimal allocations and associated cointegrating vector.

3. Having identified which countries lead EAFE, we arbitrage by the exploitation rules detailed above.

Merger Arbitrage

Mergers and acquisitions is a mainstay of financial intermediation and the global markets. Academic effort has focused on wealth effects and associated economic reasons around merger periods which is in stark contrast to traders' efforts to capitalize on mispricing preceding deal completion (Branch and Yang, 2003). Such a trading strategy is known as merger arbitrage, though it is commonly also referred to as risk arbitrage and event arbitrage.

Merger arbitrage usually involves buying the stock that is being acquired and shorting the stock of the acquirer aiming to capture the offer premium which always persists after an announcement of intention. This premium reflects the risk inherent to the deal, hence risk arbitrage, and is the difference between the offered consideration and the target price. The long-short positions reflect the spread between the offered consideration and the target price through the merger period.

To enact this, an arbitrageur would need to calculate the rate of return implicit to the current spread compared with the event risk of the deal.